ECI 149 HW 5

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1. Kinetics Practice (9 pts total):):

a) $\frac{d[HCHO]}{dt} = K_{R4}[\mathrm{CH_3O^\bullet}][O_2]$

b) $\frac{d[\mathrm{CH_3O_2^\bullet}]}{dt} = K_{R2}[\mathrm{CH_3^\bullet}][O_2][M] - K_{R3}[\mathrm{CH_3O_2^\bullet}][NO]$

c) $\frac{d[NO_2]}{dt} = K_{R3}[\mathrm{CH_3O_2^\bullet}][NO] + K_{R5}[\mathrm{HO_2^\bullet}][NO] - K_{R6}[NO_2] + K_{R8}[O_3][NO] - K_{R9}[NO_2][OH][M]$

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2. Kinetics (9 pts):

$K(t) = Ae^{-E_a/RT} \rightarrow K(293) = 1.8 \times 10^{-12}[\frac{cm^3}{molecule\ s}]e^{\frac{2.83kJ/mol)}{(8.31\times10^{-3}kJ/mol/K)(293K)}} = 5.75 \times 10^{-12}[\frac{cm^3}{molecule\ s}]$

$\frac{d[C_7H_8]}{dt} = -K[C_7H_8][OH]$

$[C_7H_8]_{STP} = \frac{1ppb\times[2.46\times10^{19}molec-air/cm^3]}{10^9} = 2.46 \times 10^{10}molec - C_7H_8/cm^3$

$\therefore \frac{d[C_7H_8]}{dt} = 5.75 \times 10^{-12}[\frac{cm^3}{molecule\ s}] \times 2.46 \times 10^{10}molec - C_7H_8/cm^3 \times 1.5 \times 10^6 molec - OH/cm^3$

$\frac{d[C_7H_8]}{dt} = 2212363 molec/cm^3/s$

Over 6 hours, 4,587,040,555 molecules of toluene will react and be lost, bringing the concentration down to $2.0 \times 10^{10} molec/cm^3$

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3. Ozone (5 pts): If a region is VOC limited, then reducing NOx will actually increase ozone levels, until you reduce NOx enough to get out of the VOC limited region. Vice-versa for a NOx limited region, except ozone won't go up if you reduce VOCs, they just won't change (to a point).

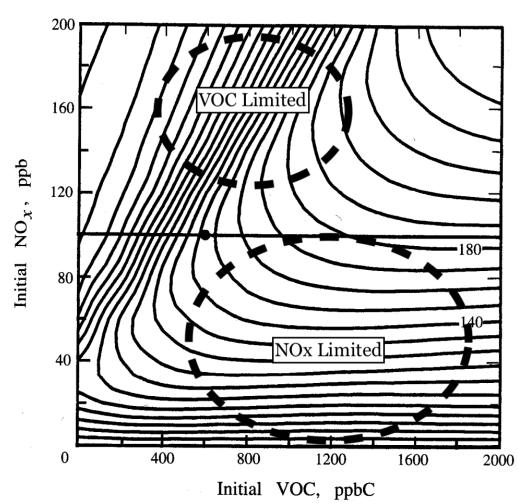

FIGURE 5.15 Ozone isopleth plot based on simulations of chemistry along air trajectories in Atlanta (Jeffries and Crouse, 1990). Each isopleth is 10 ppb higher in O_3 as one moves upward and to the right.